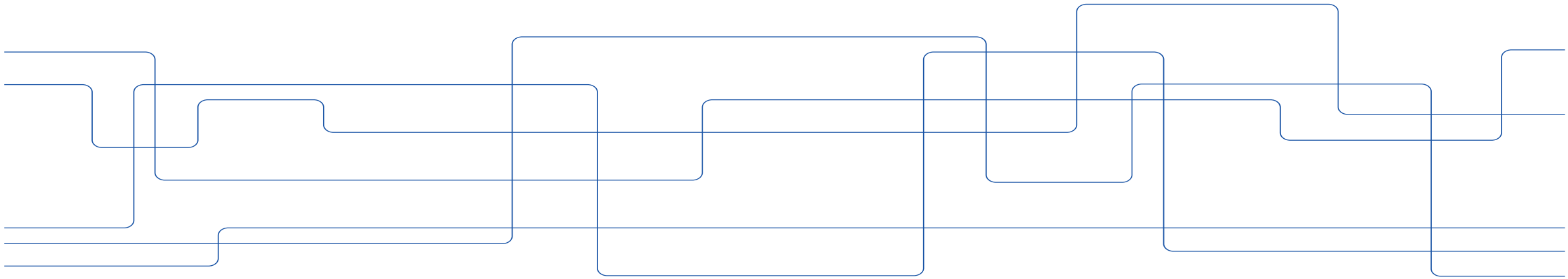
KTH ROYAL INSTITUTE OF TECHNOLOGY

Lecture 3: Bias and algorithms

Med Etik TaMos - HT2024

Barbro Fröding



Recap of key points

Recap

Different claims

- The aim of applied ethics is to provide normative guidance, i.e., *prescriptions* or *evaluations*, with regards to practical moral problems.
- Making distinctions between descriptive/predictive and evaluative/prescriptive claims

Descriptive claims: “Smith is 95 years old.”

Predictive claims: “Smith will spend her money on fried chicken and lottery tickets.”

- **Evaluative claims**: “It is good, for some, to have exoskeletons.”
- **Prescriptive claims:** “Smith should/should not be given an exoskeleton.”

Recap

Normative claims need to be justified

To justify a claim is to propose convincing reasons in favor of it.

Justifying *normative claims* involves critical reflection on reasons that carry moral relevance.

- Identify and take into account relevant facts
- Identify the *values* involved and reflect on these in light of other relevant values
- Identify and take into account possible *value conflicts*

Provide a balanced judgment with regard to all relevant reasons involved.

"Smith is 95 years old. She has a limited understanding of the world around her and cannot move much without the help of her caretakers. A new technological device, a so-called exoskeleton, could increase her autonomy as it would enable her to move around her apartment without help.

However, such **autonomy** would also enable her to, e.g., leave her building and wander off and **do things that may not be good for her**. For example, she could walk into a store and spend all her money."

Prescription: Smith should be provided with an exoskeleton with some restrictions.

Recap

Base the balanced judgment on an ethical theory

- We looked into one especially important ethical theory called principlism, which emphasizes four universal principles: **autonomy, nonmaleficence, beneficence, and justice.**

- Step 1: you need to observe which of the four principles is most relevant for the case.

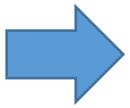
- Step 2: you need to specify them to the case

- “Specification is not a process of producing, explaining, or defending general principles, but of making them relevant and useful in specific contexts.” (Beauchamp and Rauprich 2016, 2285).

Recap

How to specify values

"Smith is 95 years old. She has a limited understanding of the world around her and cannot move much without the help of her caretakers. A new technological device, a so-called exoskeleton, could increase her autonomy as it would enable her to move around her apartment without help. However, such autonomy would also enable her to, e.g., leave her building and wander off and do things that **may not be good** for her. For example, she could walk into a store and spend all her money."



Conflict between autonomy and beneficence

Solving the conflict

- It helps if one has a theory of justification.

> "The selection of basic principles, their specification in various contexts, and solutions of moral problems by further specification or balancing are in need of **justification,** given the potential for competing judgments. Thus, principlism is not complete without **a theory of justification**, i.e., a systematic account of how to support one's judgments."
>
> (Beauchamp and Rauprich 2016)

Recap

Reflective equilibrium

- One theory of justification is reflective equilibrium

> "Principlists argue that moral judgments are justified … in conjunction with related moral and nonmoral beliefs and ultimately by the coherence of the belief system as a whole."

> "The coherence of a system of beliefs (…) is established through a reflective testing, specifying, and revising of its elements with the goal of making them as consistent, mutual supporting, and unified as possible."

(Beauchamp and Rauprich 2016)

Recap

Paternalism

- "[T]he interference of a state or an individual with another person, against their will, and defended or motivated by a claim that the person interfered with will be better off or protected from harm." (Dworkin 2014)

- Conflict between **autonomy** and **beneficence.**

- We looked at a specific example: is it morally permissible to control access to personalised risk scores?
  - Not permissible (Extreme autonomy!)
  - Permissible (Extreme beneficence!)
  - Permissible. New framework: obligations.

Recap

Medical paternalism

For an act to qualify as medical paternalism 1 & 2 below are jointly sufficient

1. a physician or other healthcare professional undertakes actions designed to limit the exercise of the patient's autonomy,
2. on the basis of clinical judgment and decision-making that are reliably beneficence based in order to protect the health-related interests of the patient from the consequences of the patient's decisions and actions based on them.

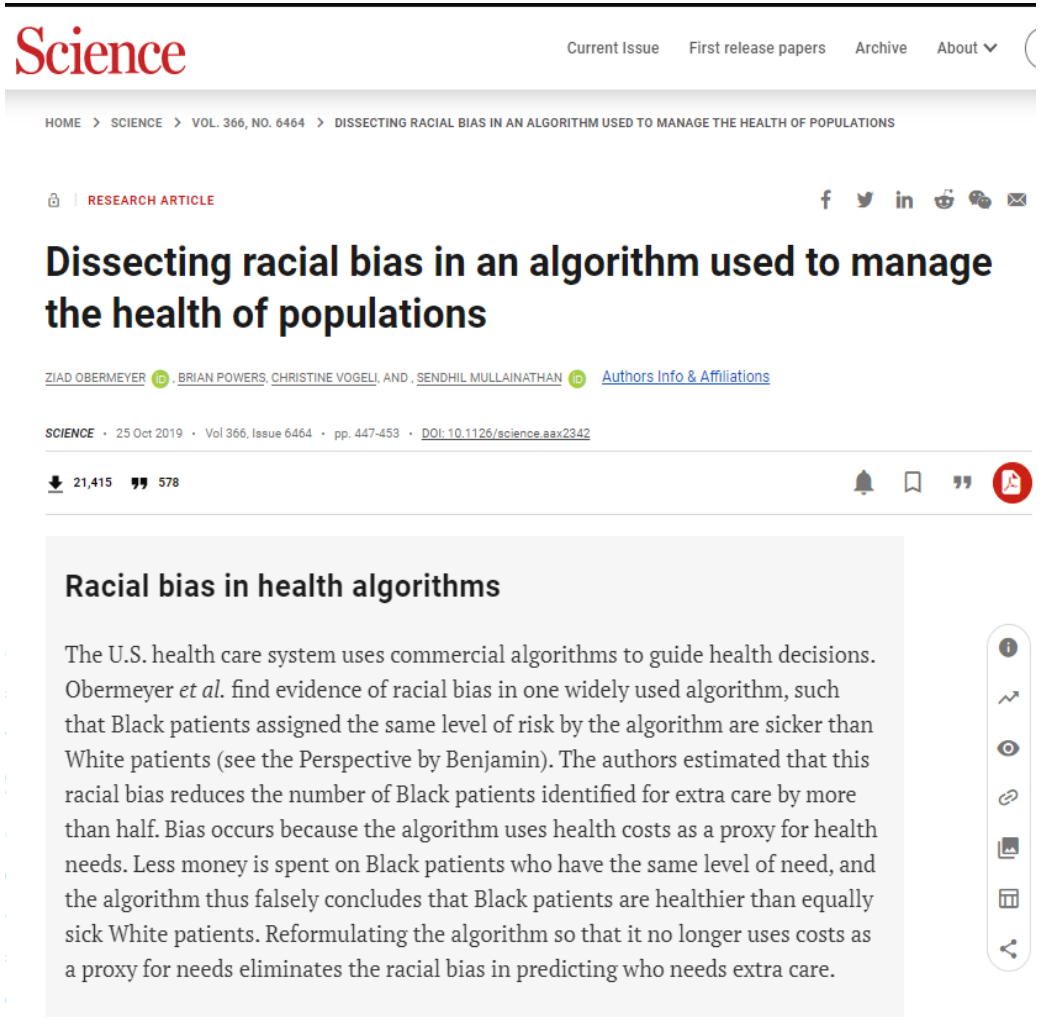
Note that *clinical judgement* in (2) is shorthand for: expert clinical judgements which are both evidence-based and beneficence-based in their evaluation of the relevant evidence.



Tip: make sure you can define medical paternalism

Bias and algorithms

An example: racial bias



Science

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RESEARCH ARTICLE

Dissecting racial bias in an algorithm used to manage the health of populations

ZIAD OBERMEYER, BRIAN POWERS, CHRISTINE VOGELI, AND SENDHIL MULLAINATHAN Authors Info & Affiliations

SCIENCE · 25 Oct 2019 · Vol 366, Issue 6464 · pp. 447-453 · DOI: 10.1126/science.aax2342

21,415 578

Racial bias in health algorithms

The U.S. health care system uses commercial algorithms to guide health decisions. Obermeyer *et al.* find evidence of racial bias in one widely used algorithm, such that Black patients assigned the same level of risk by the algorithm are sicker than White patients (see the Perspective by Benjamin). The authors estimated that this racial bias reduces the number of Black patients identified for extra care by more than half. Bias occurs because the algorithm uses health costs as a proxy for health needs. Less money is spent on Black patients who have the same level of need, and the algorithm thus falsely concludes that Black patients are healthier than equally sick White patients. Reformulating the algorithm so that it no longer uses costs as a proxy for needs eliminates the racial bias in predicting who needs extra care.

Obermeyer and co-authors gained access to the training data, algorithm and some contextual data. And they found out that the algorithmic tool used health costs as a proxy for health needs.

That means that the more money the healthcare providers spend on the health needs of patients; the more ill the algorithm thought the people were.

The whole text: Obermeyer, Z., Powers, B., Vogeli, C., & Mullainathan, S. (2019). Dissecting racial bias in an algorithm used to manage the health of populations. *Science*, *366*(6464), 447-453. https://www.science.org/doi/10.1126/science.aax2342

Problems

- Not the same amount of money is spent on people who are equally sick in the United States. (Historical, political, economic and social factors)

- As a result, for example, black patients who the algorithm assigned with the same risk score as white patients tended to be much sicker, because healthcare providers spent much less money on their care overall. (*you recall that money spent was a proxy for degree of illness*)

Assessing risk, automating racism



Racial bias in cost data leads an algorithm to underestimate health care needs of Black patients. PHOTO: FATCAMERA/GETTY IMAGES

As more organizations and industries adopt digital tools to identify risk and allocate resources, the automation of racial discrimination is a growing concern. Social scientists have been at the forefront of studying the historical, political, economic, and ethical dimensions of such tools (*1*–*3*). But most analysts do not have access to widely used proprietary algorithms and so cannot typically identify the precise mechanisms that produce disparate outcomes. On page 447 of this issue, Obermeyer *et al.* (*4*) report one of the first studies to examine the outputs and inputs of an algorithm that predicts health risk, and influences treatment, of millions of people. They found that because the tool was designed to predict the cost of care as a proxy for health needs, Black patients with the same risk score as White patients tend to be much sicker, because providers spend much less on their care overall. This study contributes greatly to a more socially conscious approach to technology development, demonstrating how a seemingly benign choice of label (that is, health cost) initiates a process with potentially life-threatening results. Whereas in a previous era, the intention to deepen racial

Benjamin, R. (2019). Assessing risk, automating racism. *Science*, *366*(6464), 421-422. DOI: 10.1126/science.aaz3873

"Data used to train automated systems are typically historic and, in the context of health care, this history entails segregated hospital facilities, racist medical curricula, and unequal insurance structures, among other factors. Yet many industries and organizations well beyond health care are incorporating automated tools, from education and banking to policing and housing, with the promise that algorithmic decisions are less biased than their human counterpart.

But human decisions comprise the data and shape the design of algorithms, now hidden by the promise of neutrality and with the power to unjustly discriminate at a much larger scale than biased individuals."

Applying principlism to a case of algorithmic bias

Which of the 4 principles apply to this case?

- non-maleficence (do no harm)
- beneficence (do good)
- justice

Specify: Justice in this context can be taken to mean, for example, ensuring equality and fairness in treatment

Construct a normative prescription, for example
“Algorithmic bias should be avoided, because it violates principles of non-maleficence, beneficence, and justice.”

Justify this normative claim with convincing reasons in favor of it. (e.g. argue why the principles are really important etc.)

But what can developers of medical technologies do to avoid generating such issues in the first place?

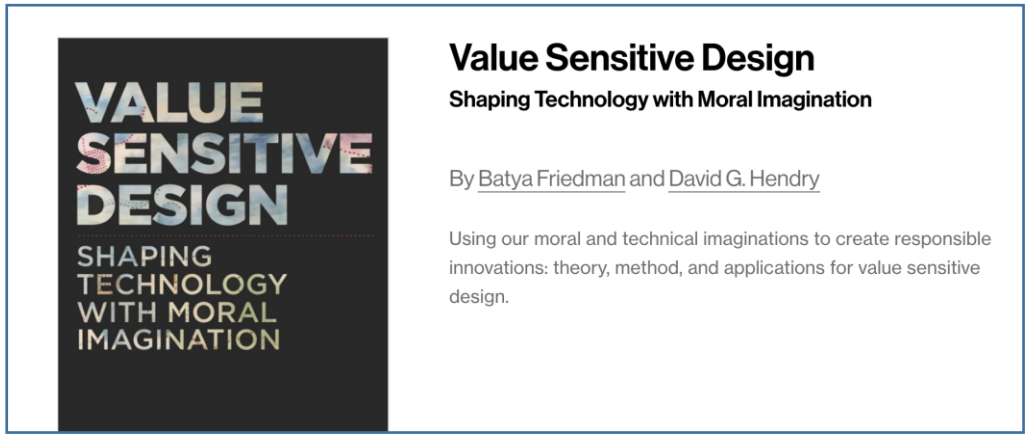


Value Sensitive Design (VSD)

Choices re. tech development and design have societal implications

” The increased awareness of possible implications of our design and our design choices for the expression (…)of our (political and) moral values contributes to an understandable desire to control and influence this process: Would it be possible to design our buildings, technologies, and institutions, so that they reflect, express, and enhance our moral and political views? ” (Manders-Huits 2011)

⇨ A method for doing just that is Value Sensitive Design





What has VSD got to do with my work as an engineer?

If you are working on medical technologies – and most other technologies – you are either actively taking part in design-decisions, or working together with designers to implement the technology in question.

Professional responsibility and codes of conduct

VSD is a useful tool for going from a theoretical discussion on ethics to concrete reflection on social impact of technology development and technology use,

The method can help you (and your team) to critically reflect on the inscription of values in the design and development stages. What values? Whose values?

- Mixed groups tends to counter bias, reduce risk of blind spots

The method can help you consider the bigger picture – placing your work in a social context: what can my work contribute, what risks attach, which values does it promote (directly and indirectly), what might the unintended consequences be, etc.?

More examples of ethically charged medical technology

Consider the increase of use of AI in healthcare, e.g.

robot surgery; medical robots (including welfare technology, rehab bots, social companion bots); image processing; decision support in diagnostics (including health apps); composition of treatment plans; online therapy and automation of clinical records.

Examples of the many ethical issues attaching e.g.

privacy; autonomy; informed consent; explainability; transparency; trust; bias; justice; security; safety, responsibility; human contacts; human dignity; human over-sight; risk of deskilling and environmental sustainability.

Currently its decision support (as opposed to autonomous)

This requires very careful design and development of the technology and an awareness of the ethical issues.

A *good* medical technology is more than a technology that works from a tech and medical point of view!

What is Value Sensitive Design?

When developing new technologies engineers tend to focus on e.g. functionality, usability, efficiency, reliability and affordability. Reflection on the ethical implications of technology design often takes place **after** it has been developed / introduced in society.

This can have very negative consequences – examples of biased and excluding design

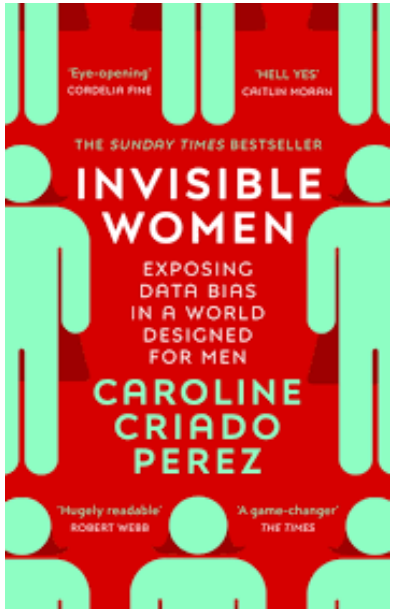
Value Sensitive Design is a methodology for proactively integrating ethics into the design and development of new technologies.

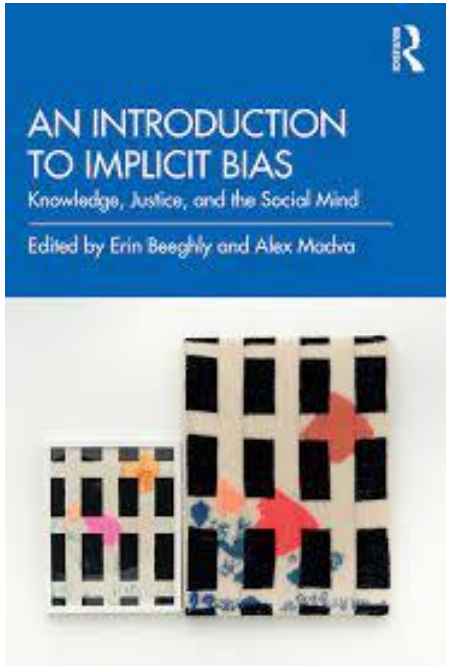
This could help to:

- reduce the risk of producing technology which undermines positive values (e.g. justice, fairness, sustainability, inclusion)
- allow us to intentionally embed a set of positive values in the technology
- promote early inclusion of stakeholders (direct and indirect) in the development and design of new technology

Further examples

- Excluding and biased design, in medicine and elsewhere in society





MIT News
ON CAMPUS AND AROUND THE WORLD

SUBS

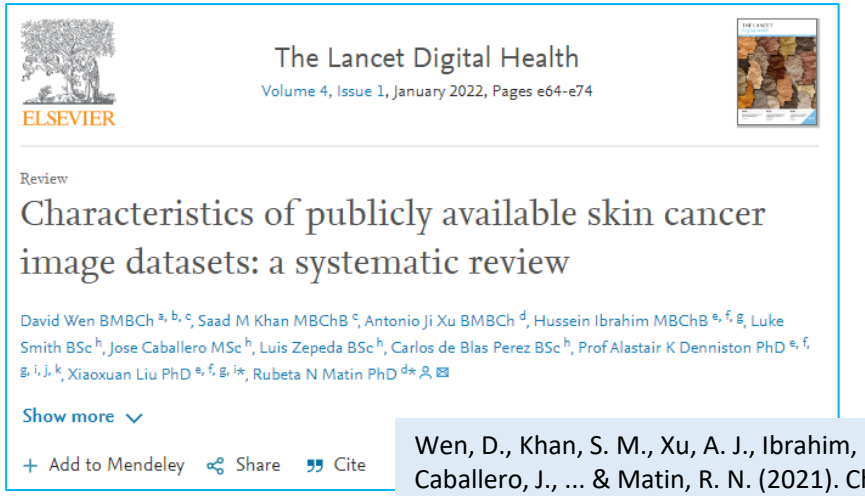
Study finds gender and skin-type bias in commercial artificial-intelligence systems

Examination of facial-analysis software shows error rate of 0.8 percent for light-skinned men, 34.7 percent for dark-skinned women.

Watch Video

Larry Hardesty | MIT News Office
February 11, 2018

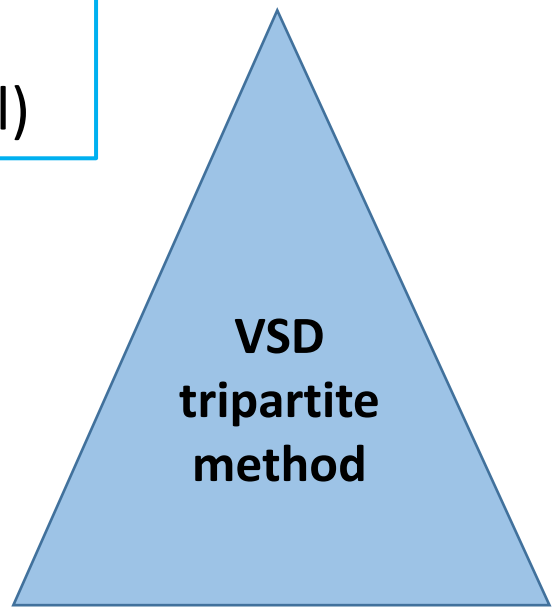
Interview with Joy Buolamwini, a researcher in the MIT Media Lab's Civic Media group. https://news.mit.edu/2018/study-finds-gender-skin-type-bias-artificial-intelligence-systems-0212



Wen, D., Khan, S. M., Xu, A. J., Ibrahim, H., Smith, L., Caballero, J., ... & Matin, R. N. (2021). Characteristics of publicly available skin cancer image datasets: a systematic review. *The Lancet Digital Health*. https://www.sciencedirect.com/science/article/pii/S2589750021002521

The research centre Data & Society studies the social implications of data-centric technologies & automation https://datasociety.net/

The methodology is an integrated, iterative, tripartite process (conceptual, empirical, technical)



Empirical investigation

- Empirically evaluate stakeholder values, and see how they relate to broader socio-cultural norms
- Then translate to potential design requirements

Conceptual investigation

- Consider values both from academic literature and from stakeholders

Technical investigation

- Technical limitations of the technology are evaluated.
- One checks how that technology supports or constraints the identified values and design requirements.

Key concepts in Value Sensitive Design

- “Within VSD, artifacts and technologies are understood as inherently morally loaded, in so far as they promote or demote specific values… already in the design and not only in the usage technologies can have moral consequences.” (Judith Simon, 2017, p. 222)

- Definition of a value in VSD
  - “…a value refers to what a person or a group consider important in life…” (Batya Friedman et al. 2013)
  - You will get some concrete examples soon

- Who are the stakeholders?
  - Direct stakeholders = people using the technology (e.g. physicians using a patient database)
  - Indirect stakeholders = people who do not use the technology but are affected by it (e.g. the patients in the database)

What values?

Examples of values to be promoted or demoted

- trust, privacy, transparency, autonomy, safety, respect for human dignity, equality, inclusion, sustainability, collaboration, diversity...

Examples of value conflicts that can occur

- privacy vs. collective security
- trust vs. security
- sustainability vs. economic development
- autonomy vs. privacy

Examples of unintended consequences

- dual use , non-intended usage, environmental effects , socio-economic effects

Common criticism of VSD

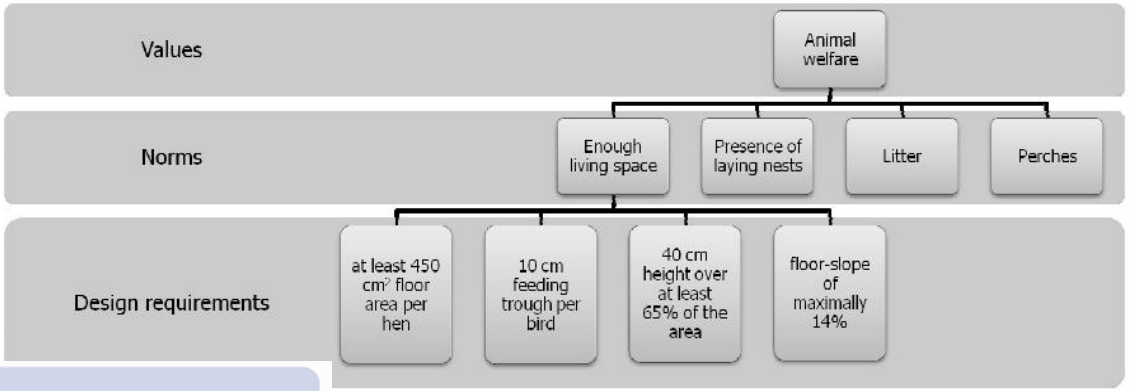
1. It's the voice of the practioners – our assessment is influenced by our own bias and values and we are not always aware of what they might be. E.g. our personal bias or, on a group level, some academic or professional communities subscribe to values which are very specific and not representative
   - One way to reduce the risk and to combat blind spots is to have mixed research teams

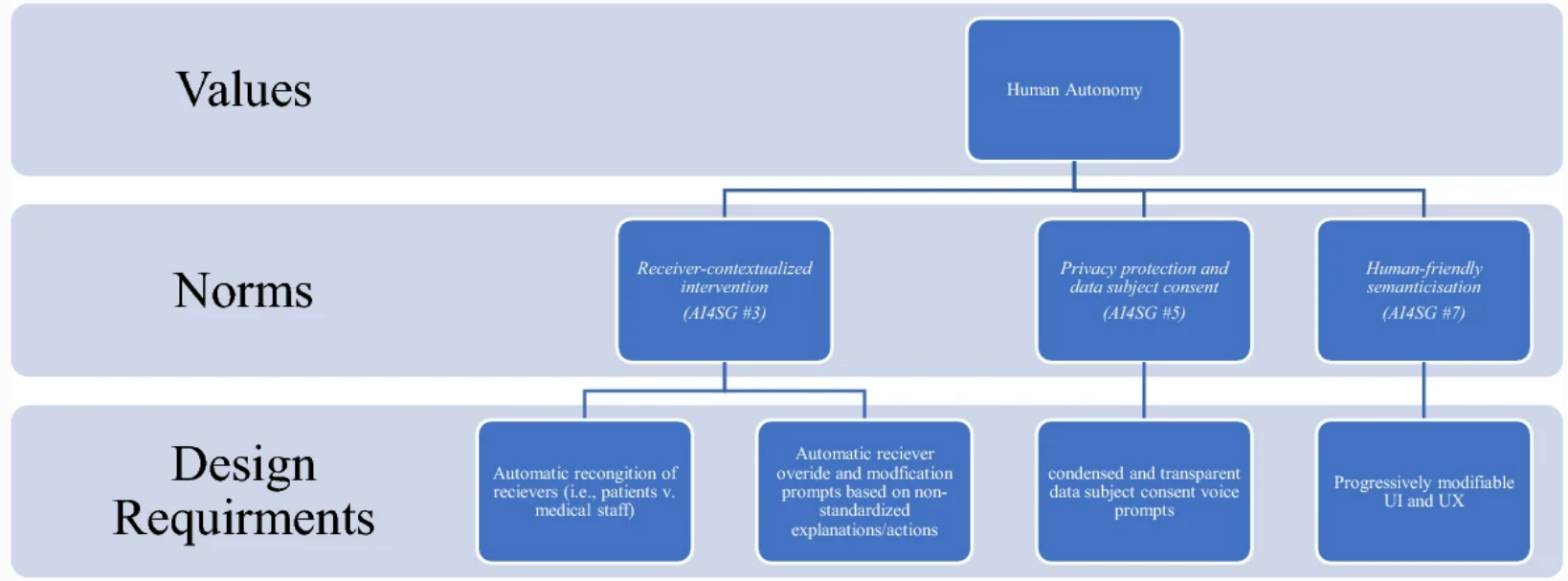
2. Is and ought – the empirical input from the stakeholders is given normative value (the normative principles should come from the underlying theory e.g. care ethics, capabilities, principlism (mid level) or perhaps virtue ethics

3. Value conflicts: not all conflicts can be easily solved e.g. a dementia patients safety vs. their privacy, the good of the individual vs the collective good.
   - To manage a trade-off between incommensurable values we need to anchor VSD in a normative theory e.g. virtue ethics.

Common criticism of VSD, cont

Complaint: VSD does not answer how to translate values into design.
Solution from the the VSD literature: value hierarchies.



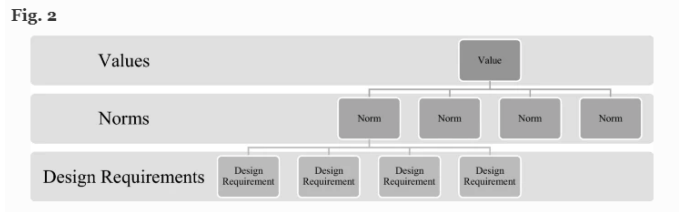


A value hierarchy which translates human autonomy to design requirements through AI4SG norms

Umbrello, S., Capasso, M., Balistreri, M., Pirni, A., & Merenda, F. (2021). Value sensitive design to achieve the UN SDGs with AI: A case of elderly care robots. *Minds and Machines*, *31*(3), 395-419. https://link.springer.com/article/10.1007/s11023-021-09561-y

For more on the idea: van de Poel, I. (2013). Translating values into design requirements. In D. P. Michelfelder, N. McCarthy, & D. E. Goldberg (Eds.), *Philosophy and engineering: Reflections on practice, principles and process* (pp. 253–266). Springer Netherlands. https://link.springer.com/chapter/10.1007/978-94-007-7762-0_20

A value hierarchy based on principlism



1. VALUE

2. NORMS

3. DESIGN REQUIREMENTS
(add more specification on technical aspects)

JUSTICE

Equal rights

Fairness (distribution)
possibly both compensating to correct past wrongs and ensuring sharing of benefits and potential costs (risks) of the new technology

Responsibility

Non-biased algorithm

Explainability

Accessibility

Affordability

Accountability

Envisioning cards – 20 cards, 4 themes

Stakeholders



Drawing from Value Sensitive Design methodology, the Stakeholder criterion emphasizes the range of effects of a technology, both on those who are in direct contact with a technology (direct stakeholders), and on those who might not be direct users, but whose lives are nevertheless affected by various interactions around the technology (indirect stakeholders).

Time



Inspired by the long-term perspective of urban planning, the Time criterion helps guide designers to consider the longer term implications of their work – implications that will only emerge after the technology has moved through initial phases of novelty to later phases of appropriation and integration into society.

Values



The Value criterion emphasizes the impact of technology on human values. Our use of the term values draws from the Value Sensitive Design literature, "what a person or group of people consider important in life." In interaction design, we have found values of interest to include but are not limited to: autonomy, community, cooperation, democratization, environmental sustainability, expression, fairness, human dignity, inclusivity and exclusivity, informed consent, justice, ownership, privacy, self-efficacy, security, trust, and universal access.

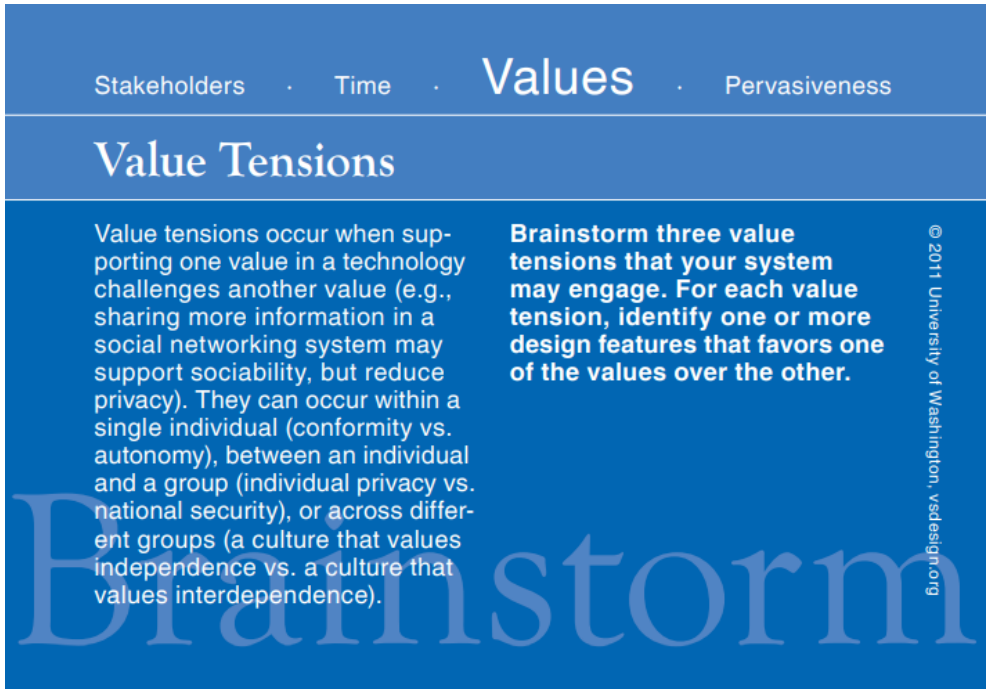
Pervasiveness



The Pervasiveness criterion emphasizes systemic interactions that follow from the widespread adoption of an interactive technology. Technologies can become pervasive with respect to geographic (e.g., city navigation software use within urban areas), cultural (e.g., text messaging within the deaf community), demographic (e.g., online social networking sites among teenagers), and other factors.

(www.envisioningcards.com)

- The case. Analyze the medical technology: social companion bots (dog/cat/seal) for elderly patients in a nursing home. The patients do not have (severe) dementia and they understand that it's a robot.
- The envisioning cards are designed to help you identify the long-term influence of new technology (this is the *conceptual* part of the VSD method).
- Take 15 min and discuss the technology through the lens of these two cards. First identify the stakeholders (e.g. patients, family, care takers, the healthcare sector, manufacturers of robots etc.), then list the key values involved; identify the potential conflicts; identify positive and negative effects on relationships. Be prepared to present to the class.





Case studies of med tech development and VSD

Digital health – last mile vaccine delivery

- *Sub-Saharan Africa: Vaccine Delivery.* The first project centers on redesigning an information system to support vaccine delivery and health worker training/supervision for 251 rural health clinics in two northern provinces in Mozambique. They used a VSD approach to engage with the stakeholders researchers identified several important values, including accountability, respect, autonomy, trust, sharing and access. The researchers subsequently chose to focus on the values of respect and accountability, by suggesting that these values affected all involved stakeholders and were considered directly relevant for the improvement of the programme.
    - Jongsma, K. R., & Jongepier, F. (2020). Value-sensitive design and global digital health. *Bulletin of the World Health Organization*, *98*(8), 579. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7411316/
    - Walton, R., & DeRenzi, B. (2009). Value-sensitive design and health care in Africa. *IEEE Transactions on Professional Communication*, *52*(4), 346-358. https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=5313650&casa_token=ewjIGenBgqgAAAAA:LYy9p4eWeScJelh7Uz0KDaRnj73MWwrznC9EfC0SirVMlZrHT1y8kbQxWqaZzg0hc55ydcmGOg

Digital health - app for preventive health care (PHC)

- *Europe*: The PHCs entail the presence for (part of) a day of students and lecturers from several disciplines of the teaching clinics in, for example, a library or neighborhood center where passersby can have a “PHC” performed, without charge and without appointment. Based on a short list of basic (health-related) questions, information about their functioning in daily life, and personal preferences, the citizen can have a more in-depth advice (sometimes including additional basic health tests) from one or more of the disciplines mentioned above. Based on the outcomes of the PHC the citizen is presented with a personal advice, for instance about lifestyle or to visit a local (health) professional. As there is no treatment relation during the PHCs, citizens will not be diagnosed or receive treatment in this context.
    - Strikwerda, L., Van Steenbergen, M., Van Gorp, A., Timmers, C., & Van Grondelle, J. (2022). The value sensitive design of a preventive health check app. *Ethics and Information Technology*, *24*(3), 1-12. https://link.springer.com/article/10.1007/s10676-022-09662-x

Case studies of med tech development and VSD, cont.

Specialized care - Physiotherapeutic Assistance System for Home Therapy

- The case deals with the design of a sensor-based assistance system to support physiotherapeutic activities at home. It supports the so-called Vojta therapy, which was developed for children with a malfunctioning central nervous system. In most cases, the parents of those children have to execute complex exercises at home on a daily basis. The project members from research (technology design, pattern recognition, micro systems) and practice (physicians, therapists, technology developer) have identified three major support potentials: 1) provision of contextual feedback and information regarding the treatment execution, 2) measurement of quality and quantity of treatment sessions, 3) support of training for apprentices.
    - Heger, O. (2019). Value Sensitive Design in Design Science Research Projects: The Cases of Affective Technology and Healthcare Technology. In *EMoWI@ Wirtschaftsinformatik* (pp. 17-26). http://ceur-ws.org/Vol-2297/paper2.pdf

Robot care for the elderly

- The values involved in the design of care robots are present across various levels of abstraction. This paper began by introducing the values that are central to care in the design of care robots for the elderly. In particular, we used the Care-Centered Value Sensitive Design (CCVSD) approach as a starting point for robotics and care ethics. We expanded on this by presenting other sources of values such as the UN SDGs, higher-level norms to be promoted by design (as much as possible), the European Commission's HLEG AI values of beneficial AI as constraining values to be designed for (as much as possible), and finally contextual values, in particular those stemming from stakeholder elicitations. These three sources of values were then translated into design requirements via norms that are specific to AI systems, i.e., AI4SG factors. In doing so, we adopted the value sensitive design (VSD) approach and modify it according to Umbrello and van de Poel's framework for a multi-tiered approach to the ethical design of AI systems. Drawing on the example of the PHAROS 2.0 socially assistive robot, this paper has provided an experimental approach for how designers can begin to direct their practices towards designing AI *for* social good.
    - Umbrello, S., & van de Poel, I. (2021). Mapping value sensitive design onto AI for social good principles. *AI and Ethics*. https://doi.org/10.1007/s43681-021-00038-3
    - Umbrello, S., Capasso, M., Balistreri, M., Pirni, A., & Merenda, F. (2021). Value sensitive design to achieve the UN SDGs with AI: A case of elderly care robots. *Minds and Machines*, *31*(3), 395-419. https://link.springer.com/article/10.1007/s11023-021-09561-y
    - van Wynsberghe, A. (2013a). Designing robots for care: Care centered value-sensitive design. *Science and Engineering Ethics, 19*(2), 407–433. https://doi.org/10.1007/s11948-011-9343-6

Summary

- VSD is a method that can help you place a given technology in a wider context and identify the social implications, e.g.
  - possible contributions
  - possible challenges / risks

- The direct / indirect stakeholders you have identified as well as the value conflicts can have implications for which partners you might want to choose.
  - Which user groups and organizations might you want to collaborate with in the development phase?

- Who / what competencies do you need on your team / among your partners in order to mitigate some of the foreseen conflicts and whom do you need to increase the chances of positive impact?

Further readings on VSD

Azenkot, S. et al. (2011) 'Enhancing Independence and Safety for Blind and Deaf-Blind Public Transit Riders' In: *Proceedings of CHI 2011*. New York: ACM Press.

Davis, J., & L. P. Nathan (2015). Value sensitive design: Applications, adaptations and critiques. In J. van den Hoven, P. E. Vermaas, & I. van de Poel (Eds.), *Handbook of ethics and values in technological design*. New York: Springer.

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